

ASSIGNMENT NO.5: Aggregate Functions and Data Analysis

Questions:

Q1) Create a MySQL database named CollegeDB and a table named StudentResult.

The table should contain the following columns:

Column Name	Data Type	Constraint
StudentID	INT PRIMARY	KEY
StudentName	VARCHAR(50)	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
FeesPaid	DECIMAL(10,2)	

Insert at least 10 records into the table.

Query-

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE StudentResult
```

```
(StudentID INT PRIMARY KEY,
```

```
StudentName VARCHAR(50),
```

```
Course VARCHAR(30),
```

```
City VARCHAR(30),
```

```
Marks INT,
```

```
FeesPaid DECIMAL(10,2)
```

```
);
```

```
INSERT INTO StudentResult
```

```
(StudentID, StudentName, Course, City, Marks, FeesPaid)
```

```
VALUES
```

```
(1, 'Aarav Sharma', 'B.Sc. Data Science', 'Pune', 85, 55000.00),
```

```
(2, 'Sneha Patil', 'B.Sc. Data Science', 'Mumbai', 78, 55000.00),
```

```
(3, 'Rahul Joshi', 'B.Sc. Data Science', 'Pune', 92, 55000.00),
```

```
(4, 'Priya Shah', 'BCA', 'Nashik', 74, 48000.00),
```

```
(5, 'Rohan Deshmukh', 'BCA', 'Pune', 81, 48000.00),
```

```
(6, 'Ananya Kulkarni', 'BCA', 'Mumbai', 69, 48000.00),
```

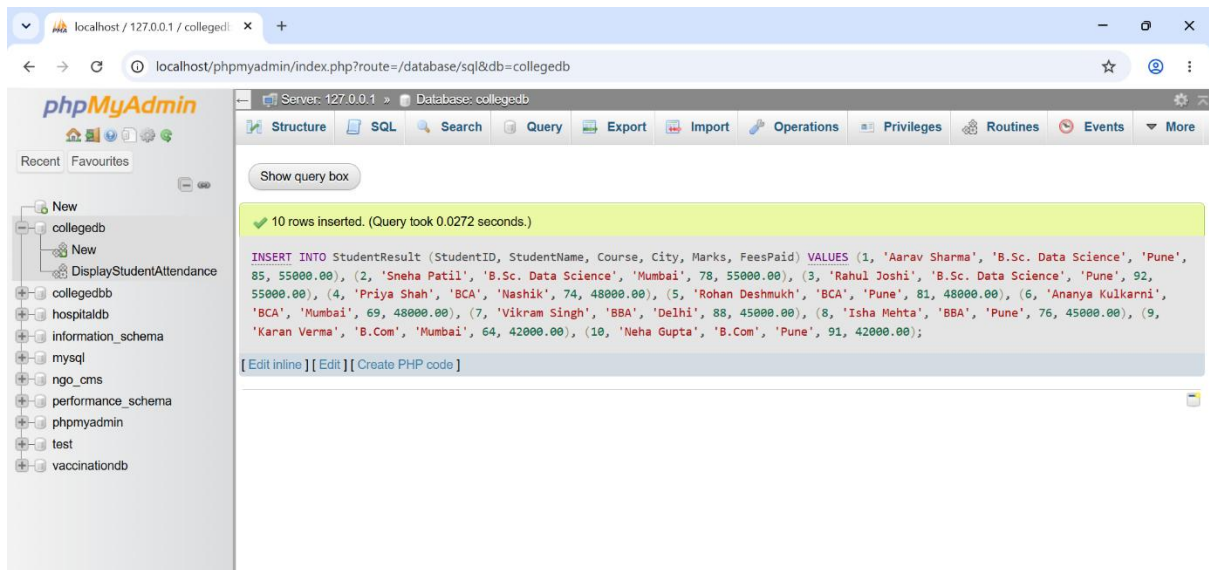
```
(7, 'Vikram Singh', 'BBA', 'Delhi', 88, 45000.00),
```

```
(8, 'Isha Mehta', 'BBA', 'Pune', 76, 45000.00),
```

```
(9, 'Karan Verma', 'B.Com', 'Mumbai', 64, 42000.00),
```

```
(10, 'Neha Gupta', 'B.Com', 'Pune', 91, 42000.00);
```

Output-



Q2) Write MySQL queries to find the following from the StudentResult table:

Requirement
Total number of students
Total fees paid
Average marks
Lowest marks
Highest marks

Query-

SELECT COUNT(*) AS TotalStudents
FROM StudentResult;

SELECT SUM(FeesPaid) AS TotalFeesPaid
FROM StudentResult;

SELECT AVG(Marks) AS AverageMarks
FROM StudentResult;

SELECT MIN(Marks) AS LowestMarks
FROM StudentResult;

SELECT MAX(Marks) AS HighestMarks
FROM StudentResult;

Output-

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

Server: 127.0.0.1 Database: collegedb Table: StudentResult

Your SQL query has been executed successfully.

```
SELECT COUNT(*) AS TotalStudents FROM StudentResult;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

TotalStudents

10

Query results operations

Print Copy to clipboard Export Display chart Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0007 seconds)

```
SELECT SUM(FeesPaid) AS TotalFeesPaid FROM StudentResult;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table

Extra options

TotalFeesPaid

483000.00

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

Server: 127.0.0.1 Database: collegedb Table: StudentResult

Showing rows 0 - 0 (1 total, Query took 0.0008 seconds)

```
SELECT AVG(Marks) AS AverageMarks FROM StudentResult;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table

Extra options

AverageMarks

79.8000

Show all Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0010 seconds)

```
SELECT MIN(Marks) AS LowestMarks FROM StudentResult;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table

Extra options

LowestMarks

64

Showing rows 0 - 0 (1 total, Query took 0.0007 seconds)

```
SELECT MAX(Marks) AS HighestMarks FROM StudentResult;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table

Extra options

HighestMarks

92

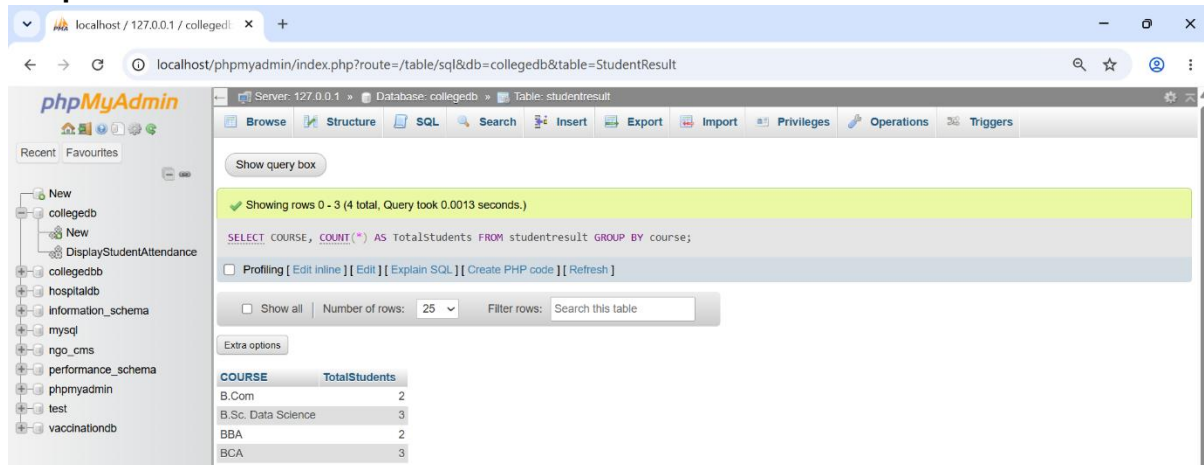
Q3) Write a MySQL query to count the number of students in each course.
The output should contain:

Column
Course
TotalStudents

Query-

```
SELECT COURSE, COUNT(*) AS TotalStudents
FROM studentresult
GROUP BY course;
```

Output-



COURSE	TotalStudents
B.Com	2
B.Sc. Data Science	3
BBA	2
BCA	3

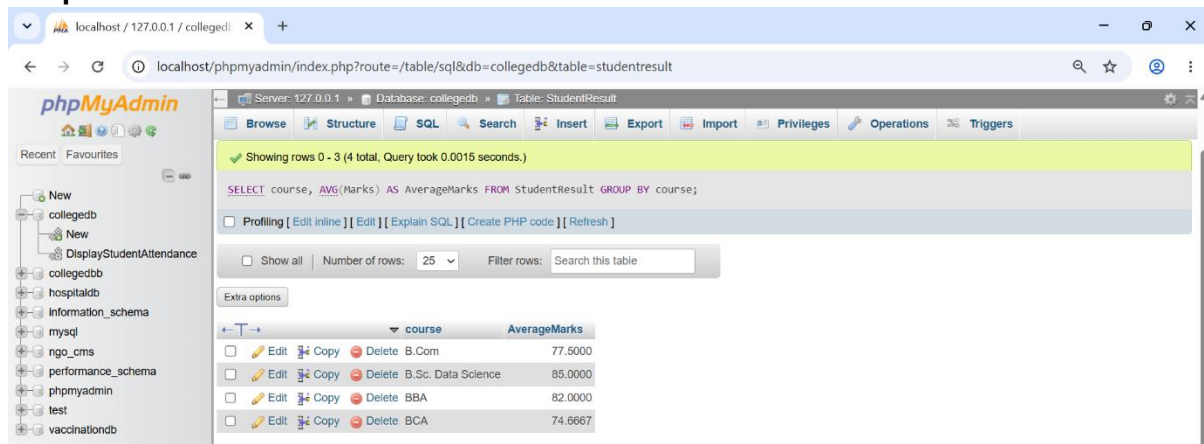
Q4) Write a MySQL query to find the average marks of students course-wise.
The output should contain:

Column
Course
AverageMarks

Query-

```
SELECT course, AVG(Marks) AS AverageMarks
FROM StudentResult
GROUP BY course;
```

Output-



course	AverageMarks
B.Com	77.5000
B.Sc. Data Science	85.0000
BBA	82.0000
BCA	74.6667

Q5) Write a MySQL query to display only those courses where the number of students is greater than 2.

Use:

Clause
GROUP BY
HAVING

Query-

```
SELECT Course, COUNT(*) AS TotalStudents
FROM StudentResult
GROUP BY course
HAVING COUNT(*) >2;
```

Output-

The screenshot shows the phpMyAdmin interface. The SQL tab is active, displaying the query: `SELECT Course, COUNT(*) AS TotalStudents FROM StudentResult GROUP BY course HAVING COUNT(*) >2;`. The status bar indicates 'Showing rows 0 - 1 (2 total, Query took 0.0025 seconds.)'. Below the query, there are options for Profiling, Edit inline, Edit, Explain SQL, Create PHP code, and Refresh. The 'Show all' checkbox is checked, and the 'Number of rows' is set to 25. The 'Filter rows' field is empty. The 'Extra options' section is expanded, showing the query results in a table:

Course	TotalStudents
B.Sc. Data Science	3
BCA	3

Q6) Write a MySQL query to analyze course performance.
The output should display:

Column
Course
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Sort the result by AverageMarks in descending order.

Query-

```
SELECT
Course,
COUNT(*) AS TotalStudents,
ROUND(AVG(Marks),2)AS AverageMarks,
MIN(Marks) AS LowestMarks,
MAX(Marks) AS HighestMarks
FROM StudentResult
GROUP BY Course
ORDER BY AverageMarks DESC;
```

Output-

localhost / 127.0.0.1 / collegedi x +

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=StudentResult

phpMyAdmin

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- vaccinationdb

Server: 127.0.0.1 » Database: collegedb » Table: StudentResult

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Show query box

Showing rows 0 - 3 (4 total, Query took 0.0014 seconds.)

```
SELECT Course, COUNT(*) AS TotalStudents, ROUND(AVG(Marks),2)AS AverageMarks, MIN(Marks) AS LowestMarks, MAX(Marks) AS HighestMarks FROM StudentResult GROUP BY Course ORDER BY AverageMarks DESC;
```

☐ Profiling [Edit Inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Course	TotalStudents	AverageMarks	LowestMarks	HighestMarks
B.Sc. Data Science	3	85.00	78	92
BBA	2	82.00	76	88
B.Com	2	77.50	64	91
BCA	3	74.67	69	81